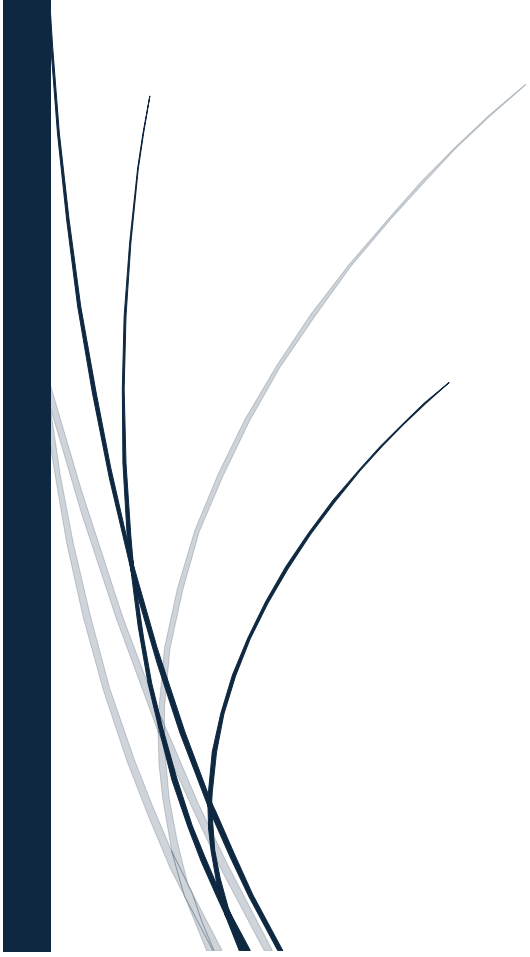
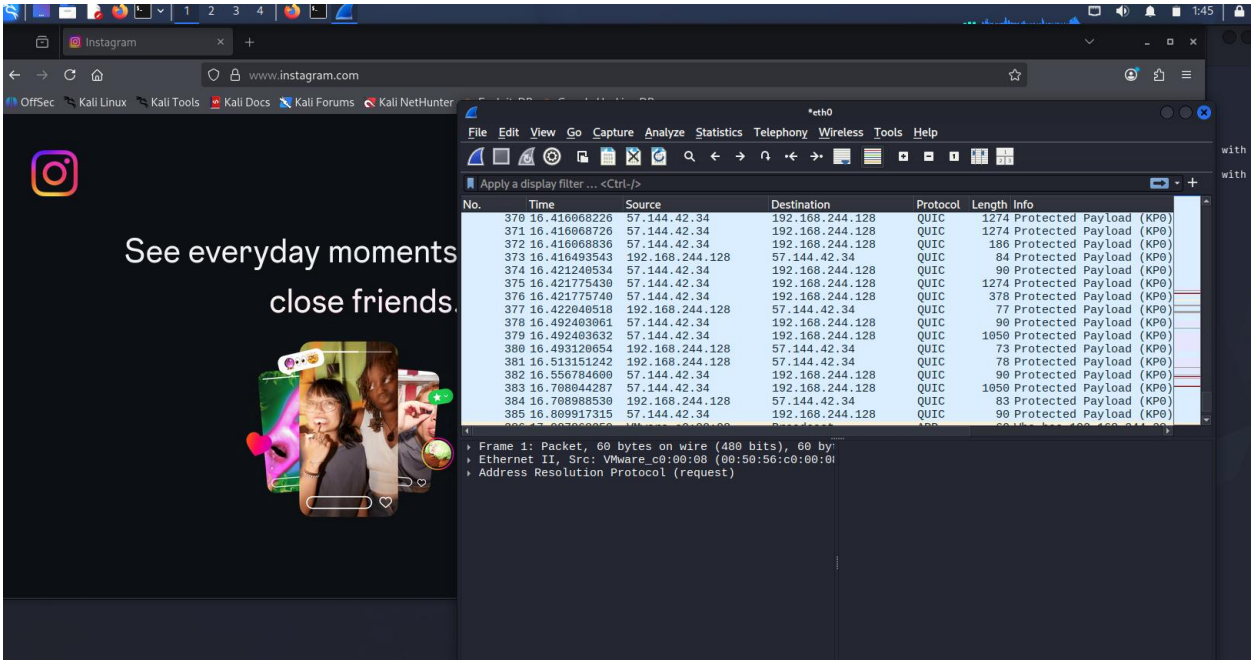
Assignment 12

Introduction to Networking & OSI Model Theory



1. **Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.**



- I opened **Wireshark** and selected my main network interface, **eth0**. Then I started capturing packets and opened the Instagram website in **Firefox** by visiting www.instagram.com**.**

- While Instagram was loading, **Wireshark captured the network** traffic between **my laptop** and **Instagram's servers.** After the page loaded, I stopped the capture and checked the packet list.

The main protocols visible in the capture were:

- **QUIC** — encrypted web traffic used by Instagram/Google Chrome-style web connections over UDP.

- **ARP (Address Resolution Protocol)** — used on the local network to find the hardware address of devices, such as the router.

➢ QUIC packets exchanged between the local IP address 192.168.244.128 and the remote IP address 57.144.42.34, which is part of the Instagram-related traffic.

2. **Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

➢ **From the Wireshark capture, I selected three packet/ protocol examples and mapped them to the OSI model:**

Packet / Protocol	**OSI Layer**	**Reason**
QUIC packet	**Transport layer (Layer 4)**	QUIC transports the encrypted Instagram web data over UDP. It handles connection and data delivery functions similar to TCP, so it is mainly associated with the Transport layer.
UDP *(inside QUIC traffic)*	**Transport layer (Layer 4)**	QUIC uses UDP as its underlying transport protocol to send packets between the laptop and Instagram's server.
ARP request	**Data Link layer (Layer 2)**	ARP is used in the local network to find the MAC address connected to an IP address, such as the router's MAC address. It works with Ethernet frames.

3. **Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).**



4. **Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint: Think about how your message is typed, packaged, sent over the internet, and received.**

Example: I type “Hello” and send it on WhatsApp.

1. **Application Layer (Layer 7)**
   type “Hello” in WhatsApp and press send.
2. **Presentation Layer (Layer 6)**
   The message is formatted and **encrypted** for secure communication.
3. **Session Layer (Layer 5)**
   It manages the **communication session** between my phone and the receiver's phone.
4. **Transport Layer (Layer 4)**
   The message is broken into **smaller parts** and prepared for delivery.
5. **Network Layer (Layer 3)**
   IP addresses are used to find the destination.
6. **Data Link Layer (Layer 2)**
   The data is put into frames and sent through Wi-Fi or Ethernet.
7. **Physical Layer (Layer 1)**
   The data travels as **radio/electrical signals** through the network.